Dijkstra’s Shortest Path Algorithm

1. Definition

Dijkstra’s Shortest Path Algorithm is a graph algorithm used to find the shortest distance from a source vertex to all other reachable vertices in a weighted graph. It was developed by Dutch computer scientist Edsger W. Dijkstra.

2. What is a Trace?

A trace is the step-by-step execution of an algorithm. In Dijkstra’s algorithm, we record each vertex’s current distance and update it whenever a shorter path is found.

3. Steps of the Algorithm

1. Select the source vertex.

2. Set the source distance to 0 and all other distances to infinity (∞).

3. Choose the unvisited vertex with the smallest tentative distance.

4. Check its adjacent vertices and update any distance that can be reduced.

5. Mark the selected vertex as visited.

6. Repeat until all reachable vertices have been visited.

4. Example Graph

Use these undirected weighted edges, with A as the source: A–B = 4, A–C = 1, C–B = 2, B–D = 1, C–D = 5, B–E = 7, and D–E = 3.

5. Step-by-Step Trace

Step	Selected vertex	A	B	C	D	E
0	Initial	0	∞	∞	∞	∞
1	A	0	4	1	∞	∞
2	C	0	3	1	6	∞
3	B	0	3	1	4	10
4	D	0	3	1	4	7
5	E	0	3	1	4	7

6. Trace Explanation

Step 1 — Select A: From A, the distance to B is 4 and to C is 1.

Step 2 — Select C: C has distance 1. Update B to 1 + 2 = 3 and D to 1 + 5 = 6.

Step 3 — Select B: B has distance 3. Update D to 3 + 1 = 4 and E to 3 + 7 = 10.

Step 4 — Select D: D has distance 4. Update E to 4 + 3 = 7.

Step 5 — Select E: No shorter distance is found.

7. Final Shortest Paths

Destination	Shortest path from A	Distance
A	A	0
B	A → C → B	3
C	A → C	1
D	A → C → B → D	4
E	A → C → B → D → E	7

8. Advantages

• Finds shortest paths efficiently.

• Useful for road maps, navigation, and network routing.

• Can be implemented with arrays, priority queues, and other data structures.

9. Limitations

• Does not work correctly with negative edge weights.

• Performance depends on the data structure used.

• Needs memory to store distances and visited vertices.

10. Applications and Conclusion

Applications include GPS navigation, computer-network routing, finding routes between cities, and transportation planning. In conclusion, Dijkstra’s algorithm finds the shortest paths from a source vertex to every reachable vertex in a weighted graph with non-negative edge weights.